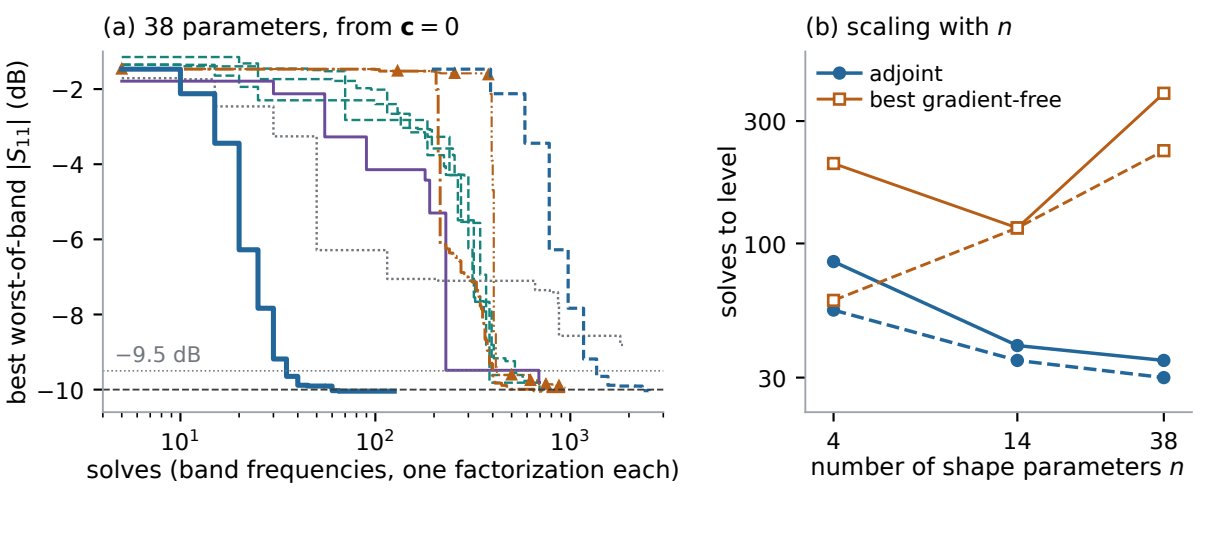
(a) 38 parameters, from $\mathbf{c} = 0$
best worst-of-band $|S_{11}|$ (dB)
−2
−4
−6
−8
−10
−9.5 dB
10^1
10^2
10^3
solves (band frequencies, one factorization each)
(b) scaling with n
adjoint
best gradient-free
solves to level
300
100
30
4
14
38
number of shape parameters n
L-BFGS, adjoint gradient
L-BFGS, finite-difference gradient
sep-CMA-ES, σ_0=0.5 (3 seeds)
CMA-ES (pycma), σ_0=0.5
sep-CMA-ES, σ_0=1.5 (1000 evals)
NEWUOA, npt 40
NEWUOA, npt 77 (Palace)